Unit 4
Computer Accessories 9 hours

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Ergonomically Designed Devices

The Input Accessories

Input Devices

Input device allows the computer system to interact with the outside world by moving data into the system i.e. an **input device** is any hardware component that allows us to enter data and instructions into a computer. Common input devices are: keyboard, mouse, joystick, etc.

Input device performs following three basic functions:

- Accepts input (data with instructions) from outside the world.
- Converts human readable data into machine readable forms.
- Transfers those data to the CPU for further processing:

Keyboard Devices:

A keyboard is a primary text input device. A **keyboard** is a device which contains keys that allow a user to enter data and instructions into the computer. All computer

keyboards have a typing area that includes the letters of the alphabet (A-Z), numbers (0-9), and other basic keys. Most input data is entered input the computer by using a keyboard. Most typewriters and computer keyboards are Qwerty Keyboards. The alphabetic keys are arranged so that the upper-left row of letters begins with the six letters QWERTY. Designers of other keyboards claim that their boards are easier to learn than the QWERTY keyboard.

A normal computer keyboard is composed of around 110 keys. Although on the basis of key arrangements there are four major layouts being used worldwide, **QWERTY, AZERTY, QWERTZ and HCESAR**, but they can be classified in many more different categories depending on types of computer connector, size, number of keys etc.

Advantages of using keyboards for data input include

- It is not necessary to buy additional equipment because most computer systems are normally supplied with keyboards.
- Entering data and instructions with keyboards is generally faster than with pointing devices.

Disadvantages of using keyboards for data input include

- It takes a lot of time to practice in order to type quickly and accurately.
- Typing speeds are still very slow when compared with computer speeds.

Point and Draw Devices

A pointing device is an input interface that allows a user to input data to a computer using physical motion - point, click, and drag. Input devices like mouse, light pen, allows a user to make rapid selection from the various icons or menus provided in the computer system but just clicking onthat menu or icons and also such devices are very effectively to create graphics elements on the screen such as line, curves or many more shapes. Such devices are also known as drawing and painting devices.

- Mouse:

The mouse is a palm-size device with a ball built into the button. The mouse is usually connected to the computer by the cable and many have from one to four buttons (usually two).

The mouse may be mechanical or optical and comes in many shapes and sizes. When you move the mouse over a smooth surface, the calls rolls, and the pointer on the display screen moves in the same direction. A **mouse** is the most widely used pointing device with a GUI environment on personal computers.

A **mechanical mouse** has a rubber ball on its underside to detect movement of the

mouse. An **optical mouse** uses devices that emit and sense light to detect the mouse's movement. A **cordless mouse** (either mechanical or optical) transmits data using wireless technology such as radio waves or infrared light waves.

Advantages:

- A mouse is user-friendly for computer beginners.
- A mouse is easy and convenient to use with a graphical user interface.
- Using a mouse to select items or move to a particular position on the screen is faster thanusing a keyboard.

Disadvantages:

- It is not easy and convenient to input text with a mouse.
- Issuing commands by using a mouse is slower than by using a keyboard.
- A mouse is not accurate enough for drawings that require high precision.
- A mouse usually requires a flat surface to operate.
- A mouse needs more desk space to operate when compared with a trackball.

- **Joystick:**

A joystick is a pointing device often used for playing games. The joystick has a gearshift-like lever that is used to move the pointer on the screed. On most joysticks, a button on the top is used to select options. In industry and manufacturing, joysticks are used to control robots. Flight simulators and other training simulators also use joysticks.

A **joystick** is a pointing device with a vertical lever mounted on a base. The lever usually includes buttons called triggers, which activate certain events when pressed. Joysticks are mainly used for computer games and ultrasound scanners in hospitals.

Yoke

A yoke is a hardware input device used with computer games, such as flight simulation games. It allows a player to fly up or down, or back and forth, by pulling or pushing the device.

The picture shows a flight yoke by Saitek and is an example of a typical yoke that connects to a computer.

- ## Touch Screen

A **touch screen** is an *input/output* touch-sensitive display. Touch screens are often used for information located in department stores, hotels, airports, and museums. Touch screens are also used for ATM machines to allow easy access of bank accounts.

- It enables the user to select an option by pressing a specific part f the screen.
- Touch screens are commonly used in information centers, fast food restaurants, railwaystations, etc.

Advantages:

- No extra peripherals are needed except the monitor.
- It allows an easy access to commands, which are usually identified by words or symbolson the screen.
- This is easiest way to enter data using the figure touch.

Disadvantages:

- Touch screens are not suitable for inputting a large amount of data because they require alot of arm movements.
- Only items already on the screen can be selected.
- A special computer monitors or monitor fitted with special hardware is necessary to providetouch screen capability

- ## Track Ball

Roller or trackballs consist of a small ball rotating freely in a fixed base (bed) containing sensors to detect rotation of the ball about two axes. These devices are sometimes incorporated in public access terminals since, unlike a mouse, they do not need a surface to work on and they are more resistant to vandals.

- ## Touch Pad:

Touch pad is a small pressure- and motion-sensitive area on a portable computer that you canuse to move the mouse pointer. When we move our finger across the pad, the mouse pointer on the screen also moves in the same direction. A touch pad is an alternative to a mouse.

The touch pad is a stationary pointing device that people find less tiring to use than a mouse or a track ball. The movement of a finger across a small touch surface is translated intocursor movement on the computer screen. The touch sensitivity surface may be just 1.5- 2-inch square, so the finger does not have to move much. Its size

makes it most suitable for thenotebooks or the laptops.

- ## Light Pen

A **light pen** is a pointing device that can detect the presence of light. Light pens are used by health care professionals (such as doctors and dentists) and design work.

A light pen is a computer input device in the form of a light-sensitive wand used in conjunction with a computer's monitor. It allows the user to point to displayed objects, or draw on the screen, in a similar way to a touch screen but with greater positional accuracy. A light pen can work with any CRT-based display, but not with LCD screens, projectors and other display devices.

The light pen became moderately popular during the early 1980s. Due to the fact that the user was required to hold his or her arm in front of the screen for long periods of time, the light pen fell out of use as a general-purpose input device.

Data Scanning Devices:

- **Bar code reader**

A **bar code reader** uses laser beams to read bar codes. A bar code reader reads a bar code byusing light patterns that pass through the bar code lines. Bar code readers are widely used in a POS e.g. supermarkets, department stores, libraries, etc.

A bar code reader works by directing a laser beam across the barcode and measuring the amount of light that is reflected back. The scanner converts the light energy into electrical energy which is then converted into data by the decoder and forwarded to a computer. They are used in almost all modern retailers, most commonly in supermarkets.

- **Magnetic Ink Character Reader (MICR)**

Magnetic **ink character recognition (MICR) reader** can read text printed with magnetized ink. Magnetic ink character recognition is used most exclusively by the banking industry for processing cheques. The characters represent the Cheque number, the bank number, and the account number of customers.

- **Image Scanner**

A **scanner** is a light-sensing input device that converts printed text and graphics into a digital formthat can be further processed by the computer. Scanners are mainly used for **image processing**, which consists of capturing, storing, analyzing, displaying,

printing, and manipulating images. Many scanners also include optical character recognition (OCR) software, which can convert an image of text into a text file that can be edited.

Two popular types of scanners are:

a) A **flatbed scanner** works like a copy machine except that it creates a file of the document rather than a paper copy.

b) A **handheld scanner** can be manually passed over the image to be scanned. A portable device you can use to copy an image from paper to your computer. Hand-held scanners are ideal for capturing small images, such as signatures and logos. A hand-held scanner is smaller, less expensive, and more portable than a flatbed scanner.

- **OCR (Optical Character Reader)**

Most **OCR readers** include a small optical scanner for reading characters and complicated OCR software for analyzing what is read.

Advantages:

- Written data and printed data can be read at the same time.
- Hard copies of documents can be read directly into a computer without retyping.
- The characters converted can later be edited by word processing software.

Disadvantages

- OCR readers often do not work well with handwritten characters or those in unusual fonts.

- **Graphics Tablet**

A **graphics tablet** is a flat, rectangular, electronic plastic board, on which a stylus writes or draws. It is mainly used for computer-aided design and drafting by architects, mapmakers, artists, and designers. Each location on the graphics tablet corresponds to a specific location on the screen. A graphics tablet can be used to digitize drawing with great accuracy.

- **OMR (Optical Mark Recognition Reader)**

Optical mark recognition (OMR) readers can sense hand-drawn marks such as small circles or rectangles made on pre-printed forms in certain places. The OMR reader first reads the answer key sheet to record the correct answers based on the patterns of light, and then reads the remaining documents and matches their patterns of light against the answer key sheet. OMR readers are often used for *marking*

multiple choice answer sheets, capturing data from questionnaires, enrolment forms, and lottery tickets.

- **Digital Camera**

A **digital camera** takes pictures and stores the images digitally, rather than on traditional film. Digital cameras use different media to store images, which include flash memory card, floppy disk and compact disc.

Advantages of a digital camera include

- Save money for buying films and developing costs in the long term.
- Images taken can be reviewed.
- Unwanted images can be deleted immediately after the preview.
- Photographic images can be digitized directly without using a scanner.

Disadvantages of a digital camera include

- Digital cameras are normally more expensive than ordinary film cameras with similar functions.
- Photo printing cost for digital cameras is generally higher than that for ordinary film cameras.

Pen Computing

A computer that utilizes an electronic pen (called a *stylus*) rather than a keyboard for input. Pen computers generally require special operating systems that support handwriting recognition so that users can write onthe screen or on a tablet instead of typing on a keyboard. Most pen computers are hand-held devices, which are too small for a full-size keyboard.

Advantages of pen computing include:

- Lightweight
- Portability
- Can be connected to home PC and data transferred onto it

Disadvantages of pen computing include:

- Stylus can be uncomfortable to use until you get used to it.

Handwriting Recognition

Handwriting recognition is a technique which allows a computer system to recognize characters and othersymbols written by hand in natural handwriting. The technology is used for identification and also on devices such as PDA and tablet PCs. After one write on a screen with a stylus, the computer turns that person's handwriting into digital text.

Advantages of handwriting recognition include:

- Handwriting recognition works regardless of which language text is written

in i.e. it is languageIndependent.

- It requires a minimal or no training to use.
- It has very minimal system requirements and provides immediate input.

Disadvantages of handwriting recognition include:

- Cannot recognize mathematical symbols
- Only understand linear structures
- It is very tiring for the arms.
- Handwriting needs to be clear for it to be recognized.

Digitizer:

Digitizer is used to create drawing and pictures using a digitizer tablet by a process called digitizing. Digitizing is a process by which graphic representations are converted into binary codes. Digitizer consists of 3 main parts:

- a flat surface called tablet,
- a small hand-held mouse-like device called puck and
- A special pen like device called stylus.

The puck is used to input existing drawings into the computer. The stylus is used to trace existing drawing placed on the tablet. The user makes contact to the tablet with stylus. As the stylus is connected to the tablet by a write, the traced image is stored in RAM and displayed on the monitor.

Microphone

A microphone is a device that translates sound vibrations in the air into electronic signals and scribes them to a recording medium or over a loudspeaker.

Microphones enable many types of audio recording devices for purposes including communications of many kinds, as well as music vocals, speech and sound recording.

Microphones can be standalone or embedded in devices such as headsets and telephones.

How do microphones work?

The most common type of microphone, the dynamic microphone, uses a coil suspended in a magnetic field that may be attached to multiple membranes for extended frequency response.

Dynamic microphones use electrical energy in the form of induction to produce the audio signal. These microphones are well suited to stage performance.

The microphone capsule contains a small diaphragm connected to a moving coil. When sound waves hit the diaphragm, it vibrates. This causes the coil to move back and forth in the magnet's field, generating an electrical current.
However, how a microphone works ultimately varies depending on its designed purpose.

One of the main considerations, aside from the type of device, is what is being recorded. The directionality of microphones is another consideration in microphone design.

Electronic Card Based Devices:

Electronic credit card makes it possible to charge online payments to one's credit card account. The details of a card can be encrypted by using the SSL (Secured Shell): Protocol in the buyer's computer, which is available in standard browsers. The most common are ATMs and POSterminals.

- **ATM:**
  Automatic Teller Machine is an interactive input/output device that enables a user to makebanking transactions from remote locations. ATMs utilize screen input as well as magnetic card readers.

- **POS** (Point of Sale): terminals are computerized cash registers that also often incorporate touch screen technology and bar-code scanners. These devices allow the input of numerous data such as item sold, price, method of payment, name or Zip code of the buyer, and so on. Some inputs are automated; others may be entered by the operators.

Speech Recognition Devices

Speech recognition or **voice recognition** is an ability of a computer, computer software program, or hardware device to decode the human voice into digitized speech that can be interpreted by the computer or hardware device. Voice recognition is commonly used to operate a device, perform commands, or write without having to operate a keyboard, mouse, or press any buttons.
Voice recognition is the process of taking the spoken word as an input to a

computer program. This process is important to virtual reality because it provides a fairly natural and sensitive way of controlling the simulation while allowing the user's hands to remain free.

Speech recognition (SR) is the translation of spoken words into text

Vision Based Devices

Vision input system are the latest input devices that can recognize the vision/image which appears in the range of its lens. It seems to be very useful and is becoming popular in different Government Departments like licensing, passport department and other authorities where personalidentification is required.

The Output Accessories

An **output device** is any piece of computer hardware equipment used to communicate the results of data processing carried out by an information processing system which converts the electronically generated information into human-readable form i.e. An output device is any peripheral device that convertsmachine-readable information into people-readable form such as a monitor, printer, plotter and voice output device.

The basic function of an output device is opposite to the input device and they are:

- Accepts data from the CPU
- Converts those machine-readable data into human readable forms.
- Transfers those data to outside the world.

Monitor

Monitors are the most popular soft-copy output devices used today. A **monitor** or a **display** is an electronic visual display for computers. The monitor comprises the display device, circuitry and an enclosure. A monitor is associated usually with a keyboard and together they form a *video display terminal (VDT).* A VDT is both an input and output devices. A user uses the keyboard to input data to a computer and the monitor to view the output from the computer.

There are two basic types of monitors: CRT monitor and LCD monitor

a) CRT monitor

It works similar to television screen and is suitable for use with non-portable computer system. A CRT monitor is a glass tube with a screen at one end and an electron gun at the other end. The ***electron gun*** of the CRT monitor includes the

electron beam generator, control grid, accelerating electrodes, and focusing electron lens. When electricity is provided to the monitor, the electron gun produces the electron beam which passes through a pair of vertical deflection plates and a pairof horizontal deflection plates and one of the horizontal plates to focus the beam at a desired point on the screen. The backside of the screen has phosphorus coated, which glows for a short time when hit by the electron beam. Since phosphor glows only for a short time (few microseconds), an image drawn on the screen would be visible only for a few microseconds and then would be fade away, unless the display mechanism draws the image repeatedly.

Advantage:

- High Refresh Rates
- Color Clarity and Depth
- Significantly lower cost
- Ability to adjust images
- Work better at multiple resolutions.
- Displays full motion video better
- Screen images viewed from side angle are as good as when viewed straight-on
- Strong glass screen can withstand routine handling, and is not easily damaged by casualtouch.

Disadvantages:

- Very Heavy and Large size limits easy and convenient movement.
- Use Large Amounts of Energy and generate Excess Heat
- High eye strain

b) LCD monitor

It is only one inch thick, light in weight, and consumes less electricity as compared to a CRT monitor. It is most suitable for use with portable computers such as laptop computers. LCD monitors use a special kind of liquid crystals to display images on the screen. The liquid crystals used in these monitors are normally transparent, but become opaque when charged with electricity.

Liquid crystal display technology works by blocking light. Specifically, an LCD is made of two pieces of polarized glass (also called substrate) that contain a liquid crystal material between them. A backlight creates light that passes through the first substrate. At the same time, electrical currents cause the liquid crystal molecules to align to allow varying levels of light to pass through to the second substrate and create the colors and images that you see

Advantages of LCD

- More usable display area than on comparably sized CRTs.
- Low frequency radiation is practically eliminated.
- Easy adjustment, storage, and movement
- Energy efficient, using only 1/3 to 1/2 the electricity of CRTs, and they don't generate heat.
- Potentially less eye strain due to reduced screen glare.
- These monitors are compact, light weight, and do not consume much desk space.
- These monitors do not consume much electricity and can even be operated by usingbatteries.
- Also, the images transmitted by these monitors do not get geometrically distorted and havelittle flicker.

Disadvantages of LCD

- Fragile screen; may result in both screen and backlight lamp damage if touched or handled.Thus, not recommended in environments where it may be handled roughly
- Contrast ratio causes darkness to not be displayed true. Darkest areas may be viewed asdark gray rather than black
- Designed only for one optimum resolution; can't adjust images.
- Best view of screen is straight-on, limiting clarity and colors of information for thoseviewing from an angle
- It is expensive

LED

Printer

A printer is an output device that produces text and graphics on paper. Printer is a device that accepts text and graphic output from a computer and transfers the information to paper, usually to standard size sheets of paper.

Impact printers

An impact printer makes contact with the paper. It usually forms the print image by pressing an inked ribbon against the paper using a hammer or pins. Following are some examples of impact printers.

Dot-Matrix Printers

The dot-matrix printer uses print heads containing from 9 to 24 pins. These pins

produce patterns of dots on the paper to form the individual characters. The 24 pin dot-matrix printer produces more dots that a 9 pin dot-matrix printer, which results in much better quality and clearer characters. The general rule is: the more pins, the clearer the letters on the paper. The pins strike the ribbon individually as the print mechanism moves across the entire print line in both directions, i.e. from left to right, then right to left,

and so on. The user can produce a color output with a dot-matrix printer (the user will change the black ribbon with a ribbon that has color stripes). Dot-matrix printers are inexpensive and typically print at speeds of 100-600 characters per second.

Daisy-wheel printers

In order to get the quality of type found on typewriters, a daisy-wheel impact printer can be used. It is called daisy-wheel printer because the print mechanism looks like a daisy; at the end of each "Petal" is a fully formed character which produces solid-line print. A hammer strikes a "petal" containing a character against the ribbon, and the character prints on the paper. Its speed is slow typically 25-55 characters per second.

Line printers

In business where enormous amount of material are printed, the character-at-a- time printers are too slow; therefore, these users need line-at-a-time printers. Line printers, or line-at-a-time printers, use special mechanism that can print a whole line at

once; they can typically print the range of 1,200 to 6,000 lines per minute. Drum, chain, and band printers are line-at-a-time printers.

Drum printer

A drum printer consists of a solid, cylindrical drum that has raised characters in bands on its surface. The number of print positions across the drum equals the number available on the page. This number typically ranges from 80-132 print positions. The drum rotates at a rapid speed. For each possible print position there is a print hammer located behind the paper. These hammers strike the paper, along the ink ribbon, against the proper character on the drum as it passes. One revolution of the drum is required to print each line. This means that all characters

on the line are not printed at exactly the same time, but the time required to print the entire line is fast enough to call them line printers. Typical speeds of drum printers are in the range of 300to 2000 lines per minute.

Chain printers

A chain printer uses a chain of print characters wrapped around two pulleys. Like the drum printer, there is one hammer for each print position. Circuitry inside the printer detects when the correct character appears at the desired print location on the page. The hammer then strikes the page, pressing the paper against a ribbon and the character located at the desired print position. An impression of the character is left on the page. The chain keeps rotating until all the required print positions on the line have filled. Then the page moves up to print the next line. Speeds of chain printers range from 400 to 2500 characters per minute.

Band printers

A band printer operates similar to chain printer except it uses a band instead of a chain and has fewer hammers. Band printer has a steel band divided into five sections of 48 characters each. The hammers on a band printer are mounted on a cartridge that moves across the paper to the appropriate positions. Characters are rotated into place and struck by the hammers. Font styles can easily be changed by replacing a band or chain.

Non-impact printers

Non-impact printers do not use a striking device to produce characters on the paper; and because these printers do not hammer against the paper they are much quieter. Following are some non- impacted printers.

Ink-jet printers

Ink-jet printers work in the same fashion as dot-matrix printers in the form images orcharacters with little dots. However, the dots are formed by tiny droplets of ink. Ink-jet printers form characters on paper by spraying ink from tiny nozzles through an electrical field that arranges the charged ink particles into characters at the rate of approximately 250 characters per second. The ink is absorbed into the paper and dries instantly. Various colors of ink can also be used.

One or more nozzles in the print head emit a steady stream of ink drops. Droplets of ink are electrically charged after leaving the nozzle. The droplets are then guided to the paper by electrically charged deflecting plates [one plate has positive charge (upper plate) and the other has negative charge (lover plate)]. A nozzle for black ink may be all that's needed to print text, but full-color printing is also possible with the addition of needed to print text, but full-color printing is also possible with the addition three extra nozzles for the cyan, magenta, and yellow primary colors. If a droplet isn't needed for the character or image being formed, it is recycled back to its input nozzle.

Several manufacturers produce color ink-jet printer. Some of these printers come with all their color inks in a cartridge; if you want to replace on color, you must replace all the colors. Other color ink-jet printers allow you to replace ink individually. These printers are a better choice if user uses one color more than other colors. These printers produce less noise and print in better quality with greater speed.

Laser printers

A laser printer works like a photocopy machine. Laser printers produce images on paper by directing a laser beam at a mirror which bounces the beam onto a drum. The drum has a special coating on it to which toner (an ink powder) sticks. Using patterns of small dots, a laser beam conveys information from the computer to a positively charged drum to become neutralized. From all those areas of drum which become neutralized, the toner detaches. As the paper rolls by the drum, the toner is transferred to the paper printing the letters or other graphics on the paper. A hot roller bonds the toner to the paper.

Laser printers use buffers that store an entire page at a time. When a whole page is

loaded, it will be printed. The speed of laser printers is high and they print quietly without producing much noise. Many home-use laser printers can print eight pages per minute, but faster and print approximately 21,000 lines per minute, or 437 pages per minute if each page contains 48 lines. When high speed laser printers were introduced they were expensive. Developments in the last few years have provided relatively low-cost laser printers for use in small businesses.

Advantages of Laser Printer

- The main advantage of Laser printer is its speed & efficiency at which it printshigh-quality quality graphics & text.
- Laser printers produce high-quality output as compared to other printers.
- Laser printers are quite and does not produce disturbing sounds.
- They are also capable to produce color prints.

Disadvantages of Laser Printer

- The main disadvantage of Laser printer is its cost, they are relatively costly ascompared to other printers.
- The maintenance, repair & servicing charges are also high of these printers.
- Laser printers emit small amount of ozone and are hazardous to health and theatmosphere.

Quality of printing

Quality of printed output is directly proportional to resolution of printing, measured in dots per inch (dpi). The dpi qualifier refers to the capability of a printer to print number of dots per linear inch, horizontally or vertically i.e. a 300-dpi printer is capable of printing 90,000 (300 x 300) dots per square inch.

According to this classification, printers can be classified as: *Near-letter quality (NLQ), letter quality (LQ), near-typeset quality (NTQ)* and *typeset-quality (TQ)*

A computer output device that draws graphs or pictures, usually by moving a pen.

Plotter

Plotter is an importantoutput device used to print high quality graphics and drawings. Although the graphics can be printed on printers, the resolution of such printing is limited on printers. Plotters are generally used for printing/drawing graphical images such as charts, drawings, maps of engineering and scientific application

A plotter is a printer that interprets commands from a computer to make line drawings on paper with one or more automated pens. Unlike a regular printer, the plotter can

draw continuous point-to-point lines directly from vector graphics files or commands. Some important types of plotters are:

a) **Flat Bed Plotters:**

These plotters print the graphical images by moving the pen on stationary flat surface material. They produce very accurate drawings.

A flatbed plotter is also known as **Table Plotter**. It plots on paper that is spread and fixed over a rectangular flatbed table. The flatbed plotter uses two robotic drawing arms, each of which holds a set of colored ink pens or pencils. The drawing arms move over the stationary paper and draw the graph on the paper. Typically, the plot size is equal to the area of a bed.

b) **Drum plotters:**

These plotters print graphical images by moving both the pen and the drum having paper. They donot produce as accurate drawings as printed by flatbed plotters.

A drum plotter is also known as **Roller Plotter**. It consists of a drum or roller on which a paper is placed and the drum rotates back and forth to produce the graph on the paper. It also consists of mechanical device known as Robotic Drawing Arm that holds a set of colored ink pens or pencils. The Robotic Drawing Arm moves side to side as the paper are rolled back and forth through the roller. In this way, a perfect graph or map is created on the paper. This work is done under the control of computer. Drum Plotters are used to produce continuous output, such as plottingearthquake activity.

c) **Inkjet plotters:**

These plotters use inkjet in place of pens. They are faster than flatbed plotters and can print multi- colored large drawings

Unlike a traditional inkjet printer (where dots are created from spraying dots on the paper) a plotter prints using a movable head that holds a pen structure. The pen moves back and forth as thepaper moves through the feed. The plotter creates lines that can be used in images or graphs. However, the plotter is made for drawing lines, so using it for large images takes longer than a traditional printer.

GPS

<<Assignment>>

Screen Image Projector

An image projector is an optical device that projects an image (or moving images) onto a surface, commonly a projection screen so that a group of people can view it simultaneously. It is very useful for making presentation to a group of people with direct use of a computer.

Screen image projectors have become common presentation equipment today. It provides a temporary, soft-copy output as like monitors.

Headphones

<<Assignment>>

Soundcard/video Card

<<Assignment>>

Speaker

<<Assignment>>

Voice Response Systems

Voice recognition is "the technology by which sounds, words or phrases spoken by humans are converted
into electrical signals and these signals are transformed into coding patterns to which meaning has been assigned".

Voice recognition is the process of taking the spoken word as an input to a computer program. This process is important to virtual reality because it provides a fairly natural and intuitive way of controlling the simulation while allowing the user's hands to remain free.

Besides, voice recognition programs are: either **discrete** or **continuous**. **Discrete speech** means that the speaker has to speak slowly and separate each word with a short pause. **Continuous speech** means that the speaker may speak in a flowing conversational tone

Advantages of voice recognition include

- No typing of data is necessary.
- The system can be used remotely by telephone or by people whose hands are

occupied or disabled.

- Voice recognition systems are also ideal for blind or visually impaired users

Disadvantages of voice recognition include

- Error rate is still high at the moment.
- Recognition of words is slow.
- Words sound the same (e.g. see and sea) cannot be distinguished.
- The system is not suitable for use in noisy places.
- The software must be trained to recognize specialist or technical words.
- Many people find it difficult to speak in ‘writing’ style

Computer Output Microfilm (COM)

Computer output microfilm or "COM" is a process for copying and printing data onto microfilm (A film bearing a photographic record on a reduced scale of printed or other graphic matter.) from electronic media found on personal, mini, or mainframe computers.

COM consists of:

- A high-speed recorder that transfers digital data onto microfilm using laser technology; and
- A processor that develops the microfilm once exposed to a light source.

A COM recorder can operate “on-line” or “off-line,” meaning that it can be connected to a single computer, a local or wide-area network, a minicomputer, or a mainframe computer. In addition, the recorder can operate independently as a stand-alone device that reads digital data from formatted magnetic media, such as tape. A COM recorder generally operates with:

- A duplicator that generates copies of microfiche; and
- A sorter that separates duplicate microfiche cards into bins for easy, end-point distribution.

Each functional unit is connected to the next, providing a linear path from creation to end-point distribution of the microfiche.

Computer Output Microfilm systems are still used today, mostly by organizations that need to store payroll, accounting, insurance, inventory, or employee data.

COM to CD Service

MIS can also arrange for Compact Disc transfer service. Data that has been transferred to COM for retention purposes can be transferred to CD-ROM for ease of access and very high storage capability.

The cost of this service varies based on the number of frames recorded to CD. For more information or pricing for COM to CD service, contact MIS.

Benefits of COM

COM offers many advantages that assist agencies with efficient office operations and adherence to sound records management principles. Key benefits include:

a) Reduction of Paper

One of the primary objectives of providing COM is to decrease paper use as allowed by Public Law 40 - 1979, Section 16, which authorizes recording, copying, and reproducing records by photo static, photographic, or micrographic process to reduce storage space. A one-cubic-foot box of paper records holds an average of only 3,000 pages. One microfiche card holds 230 documents (or images), and a one-cubic-foot records storage box holds 6,000 microfiche cards- a minimum of 1,380,000 pages. To

store that many paper documents, 460 one-cubic-foot records storage boxes would be needed.

b) Cost Reduction

Using COM to store or distribute information is more economical than most electronic media, and is even less expensive than paper. The cost of printing a standard 8”x11” page from a centralized printer is approximately three cents per sheet, versus only 0.0033 cents per sheet for COM microfiche. Other cost savings can be realized through the decrease in office and warehouse space needed to store paper documents.

c) Improved Quality

The newest COM technology provides superior image quality for improved user productivity. Such features as enhanced titling, enlarged file breaks, and bar coding make access and filing easier. Improved quality also means reader and printer copies that are legible.

d) Improved Service

The use of COM services outsourced by MIS helps reduce the risk of lost tapes and slow turn-around times from individually-contracted vendors, thereby offering agencies quick access to high-quality microfilm images. Initial transfer of information from the agency to the Micrographics lab is also simple and quick: database (ASCII or EBCIDIC) or image (TIFF Group IV) files may be sent directly from your agency's computers to the Micrographics FTP site for transfer to fiche. Contact the Micrographics lab for specific file-type requirements and FTP address.

e) Electronic Record Retention/Archiving

With the proliferation of electronic records, and with few rules governing the

purchase and use of imaging systems, it is vitally important that state agencies have the means to archive records from a variety of electronic sources. Using COM, electronic records can be moved from diverse, incompatible electronic storage systems to a “universal” reader.

SGD Speech Generation Device
<<Assignment>>

Ergonomically designed systems

Ergonomics is the science that deals with designing and arranging machines and other things in such a way that their users can use them with maximum comfort. This is important for achieving maximum efficiency of the people working on these machines and protecting them from accidents or medical problems caused due to continuous usage of a machine for long hours.

The word **ergonomic** is the study of the interaction between individuals, machines or appliances in their environment that can affect their performance and productivity.

The purpose of Ergonomic is to assess the interaction, or relationship between each component for determining optimum improvement in performance. Ergonomic therefore examines the relationship between three elements: the individual, the equipment, the environment and how they can beimproved.

Ergonomic is usually the focus of work as in employment performance, largely in relation to the operation of machines /equipment in the employment environment.

For example: prolonged usage of a badly designed keyboard can cause wrist problems, sometimes even serious as to require surgery. Similarly, prolonged usage of badly designed monitors can cause eyesight problem. To prevent such types of problems, manufacturers of I/O devices are working continuously towards making ergonomically better devices.

Some examples of ergonomically designed device are:

- Many new keyboards have slope adjustment and hand-rest pads to make their use more comfortable for the operators.
- A cordless mouse is more comfortable to use than a wired mouse.
- Flat screen monitors are fast replacing curved-screen monitors because they improvereadability and reduce reflections.

- Color monitors are fast replacing monochrome monitors because they are more comfortableto view.
- Newer monitors use higher resolution to prevent flicker or displayed objects, which may leadto eye problems.